A Security Enhanced Blockchain System Supporting Standard Elliptic Curve Digital Signature Algorithm Signature (ECDSA) and SM2 Signature Simultaneously

Jing Yang
Chengdu Aircraft Industry (Group) Co., Ltd.
yangjing.cf@cf.com

Yunfu Tang †
Chengdu Aircraft Industry (Group) Co., Ltd.
tangyunfu.cf@cf.com

Qunbo Zhou†
Chengdu Aircraft Industry (Group) Co., Ltd.
zhouqunbo.cf@cf.com

Chuanjie Jin †
University of Electronic science and Technology of China
Jinchuanjie.uestc@gmail.com

ABSTRACT

Blockchain provides a promising way that could ensure the traceability of transactions as well as provide strong trust establishment of users. Considering that digital signatures are a typical way to achieve mutual authentication, Elliptic Curve Digital Signature Algorithm (ECDSA) is implemented in most blockchain platforms such as Bitcoin, Etherum and Hyperleger Fabric since its superiority in security and efficiency. In 2010, the Chinese State Cryptography Administration published SM2 algorithm as a standard for commercial applications. The security, efficiency and practicality have been proven and recognized. Therefore, to improve the security and computing efficiency of the blockchain system, this paper proposed a blockchain system that supports the ECDSA algorithm and SM2 algorithm simultaneously. Each blockchain node or payer/payee is able to generate and verify both SM2 signature and ECDSA signature. we also analyze the performance and security of ECDSA and ECDSA-SM2.

CCS CONCEPTS

• Security and privacy • Cryptography • Public key (asymmetric) techniques • Digital signatures

KEYWORDS

Blockchain, Bitcoin, Digital signature, ECDSA, SM2

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1 Introduction

Bitocin is a decentralized application to transfer cryptocurrency [1] that records all transactions in a distributed append-only public ledger named blockchain [2]. Since the whitepaper of Biocoin form Nakamoto published [3] in 2009, due to the key characteristics of the blockchain, such as decentralisation, persistency, anonymity, auditability, etc [4], Bitcoin has attracted a lot of attention from both academia and industry. In the transaction of Bitcoin, a payer initiates a transaction and sends some bitcoins to the payee via broadcasting digitally signed transactions to the blockchain network [5]. The participating nodes in this peer-to-peer (P2P) network known as miners will verify the timestamp and signatures. Elliptic Curve Digital Signature Algorithm (ECDSA) has been adopted in most blockchain platforms such as Bitcoin, Ethereum [6] [7] and Hyperledger Fabric [8] [9]. Besides Bitcoin, blockchain could be applied to diverse applications far beyond cryptocurrencies [10]. Since there is no central bank or any intermediary in blockchain, blockchain is suitable to be used in financial services such as online payment and digital assets [11]. Meanwhile, blockchain has also been largely used in internet of things (IoT) [12] [13], internet of vehicles (IoV) [14] [15], healthcare [16] [17], industries [18] [19] [20], etc.

In 1991, U.S. National Institute of Standards and Technology (NIST) published the first official digital signature standard DSA signature scheme [21] that relies on the discrete logarithm problem (DLP) over finite fields. ECDSA is the elliptic curve version of DSA [22] [23], which achieves the same security with relatively smaller key size and system parameters. Currently, most applications adopt digital certificates achieved by RSA public key algorithms to tackle the authentication problems in the network [24]. Although RSA public algorithm is widely used and simple, the unbalanced encryption and decryption operation brings a lot of challenges. Meanwhile, with the big integer factorization

algorithm is being persistently improved [25], computers calculating speed is improving and the emergence of cloud computing [26], the length of the RSA key should be increased to ensure the security of the algorithm, which also increase the overhead of decryption and other operations. Compared with the RSA public key algorithms, Elliptic Curve cryptosystem ECC (Elliptic Curve Cryptography) achieves higher safety against the attack with the same key length [27] [28]. In other words, the overhead of the key pair of ECC algorithms is lower than that of the RSA algorithm. In addition, the computation efficiency of ECC algorithms is also better than that of the RSA algorithm [29].

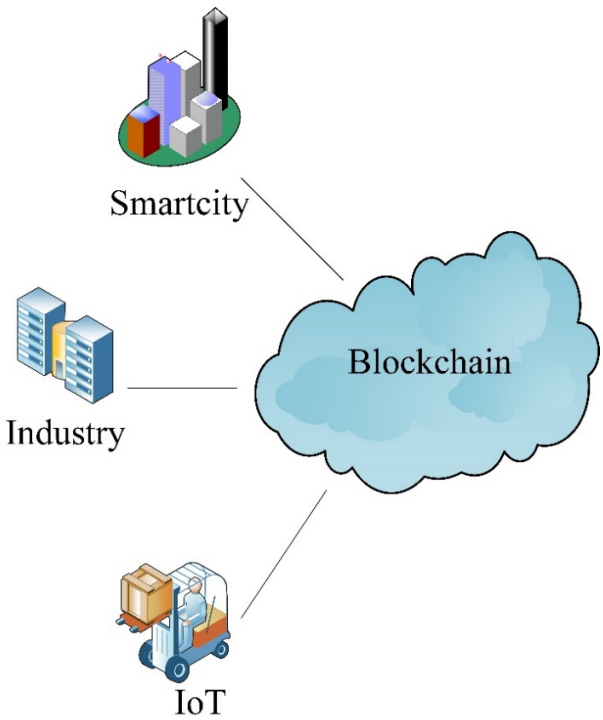

Figure 1: The adoption of blockchain

In 2010, Chinese State Cryptography Administration published SM2 [30] that is the public-key cryptosystem based on elliptic curve as a standard for commercial applications. Compared with the standard ECC algorithm, the SM2 algorithm improves the algorithm complexity, the process of encrypting messages, the structure of the encrypted messages and the form of data to be signed. Hence, the SM2 algorithm has advantages in terms of the security and the authentication of ciphertext [31]. The SM2 digital signature algorithm is one of the essential part of SM2 standard that relies on the elliptic curve discrete logarithm problem (ECDLP) [32]. Considering the security and efficiency advantages of the SM2 algorithm, a large number of companies or organizations tend to use the SM2 algorithm to replace the original digital signature algorithm such as ECDSA.

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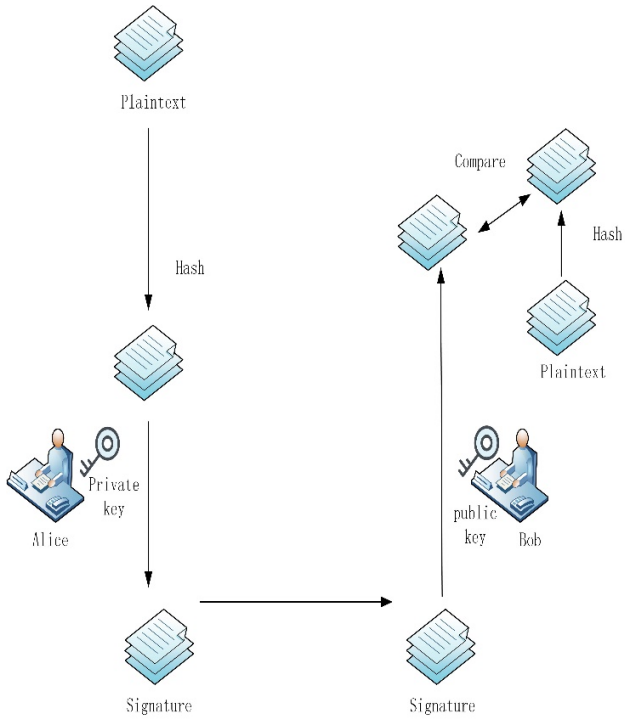

Figure 2: Digital signature

1.1 Contribution

In order to implement cryptographic algorithms that comply with national standards and improve the security and computing efficiency of the blockchain system, this paper proposes a blockchain system that supports the ECDSA algorithm and SM2 algorithm simultaneously. Each blockchain node or payer/payee is able to generate and verify both SM2 signature and ECDSA signature. The performance and security of ECDSA and ECDSA-SM2 is also analyzed.

1.2 Organization

Section 2 introduces the mathematical definitions ECDLP and a brief introduction of SM2. Section 3 describes the construction of our proposed system that supports standard ECDSA signature and SM2 signature simultaneously. The efficiency analysis is given in Section 4. Besides, Section 5 summarizes this paper.

2 Preliminaries

2.1 ECDLP

Elliptic curve cryptosystem is based on elliptic curve discrete logarithm problem (ECDLP) for their security.

Assumed that P and Q are points of the elliptic curve while Q = kP denotes the point addition operation of P for k times. If k is large enough, it is computationally infeasible to learn k.

2.1 SM2

In 2010, Chinese State Cryptography Administration published SM2 [30] that is the public-key cryptosystem based on elliptic curve as a standard for commercial applications. To be specific, SM2 is able to be applied for data encryption and decryption, key-exchanging, as well as digital signature andverification. The vital functions to implement SM2 algorithm is described as follows:

1) Big number arithmetic functions: The functions of big number arithmetic functions is the base stone of SM2. OpenSSL, Botan and GNU MP (GMP) Bignum Library are often used to implement the big number arithmetic functions.

2) Prime modular functions: To implement Sm2, modular square, modular inversion and modular multiplication operations are required.

3) Elliptic curve functions: the elliptic curve point scalar multiplication is the basic calculation of Elliptic Curve points.

4) Random number generation: To generate a secure random number for SM2, a typical way is to collect information about the current environment and generate a seed with high entropy. Then, execute pseudo-random number generator (PRNG) to extend the seed to a random number.

3 Construction

This section describes the construction of the proposed protocol that supports ECDSA and SM2-ECDSA simultaneously. Each node in the blockchain is able to generate and verify ECDSA and SM2-ECDSA. The architecture of the proposed system is described as Fig 3.

3.1 Generate and Verify ECDSA

3.1.1 Generate keys of ECDSA

The parameters of the elliptic curve digital signature algorithm (ECDSA) that common to all nodes in the blockchain is listed as follows:

1) q is the order of the prime field Fq.

2) E: $y = x^2 + ax + b$ is an elliptic curve defined over the prime field Fq.

3) P is a random non-zero base point in E(Fq).

4) n denotes a prime that is the order of P.

5) h denotes the selected hash function.

In ECDSA, MD5, SHA-1, SHA-224, SHA-256, SHA-384 and SHA-512 are all avainable and permitted. However, the hash algorithm used in ECDSA-SM2 should be SM3.

The bit string output by SM3 is 256 bits.

The security of SM3 is basically the same as SHA-256, but higher than MD5, SHA-1, SHA-224.

3.1.2 Generate ECDSA

For generating the standard ECDSA signature (r, s) on a message m, the first phase is to generate the public key Q and secret key d. The system will perform the following steps:

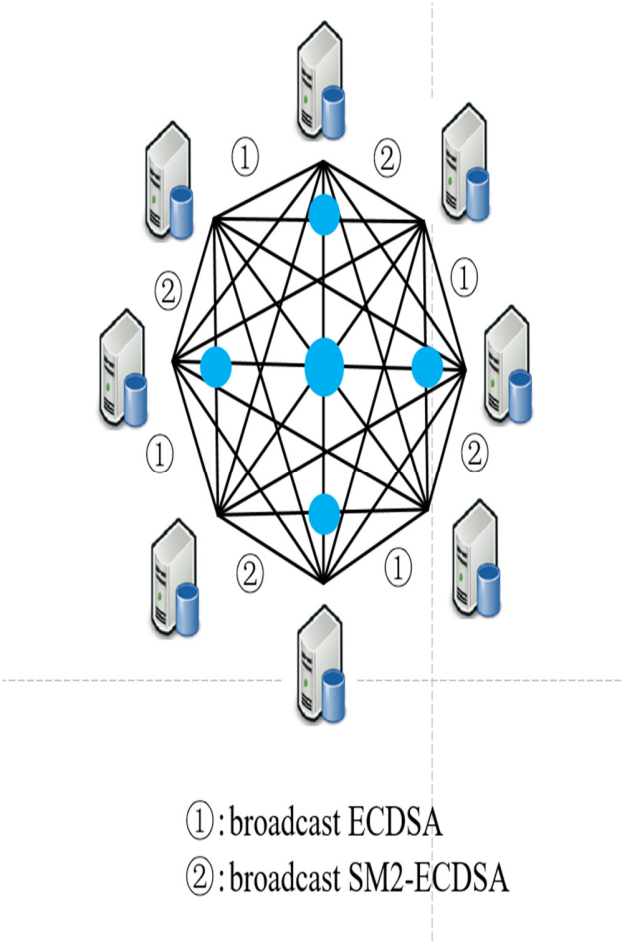

Figure 3 The architecture of the system

1) Select a random number d $\in$ [1, ..., n − 1].

2) Compute $Q = dp$.

3) Return (Q, d), where Q is the public key and d is the secret key.

Then, generate the standard ECDSA signature using the secret key d:

1) Compute the hash value of message m as: e = h(m). h is the selected hash function.

2) Select a random number k $\in$ [1, ..., n − 1].

3) Compute $R = (x, y) = kP$.

4) r = x mod n, where x is the x-coordinate of R in the above steps.

5) $s = k^{-1}(e + dr)(\mathrm{mod}\ n)$.

6) Return (r, s) as the signature.

3.1.3 Verify ECDSA

To verify the standard ECDSA signature (r_0, s_0) on a message m0, the system will perform the following steps:

1) Compute $w = s_0^{-1}(\mathrm{mod}\ n)$.

2) Compute $u = h(m_0)w(\mathrm{mod}\ n)$.

3) Compute $v = r_0 w(\mathrm{mod}\ n)$.

4) Compute $R = (x_1, y_1) = uP + vQ$.

5) If $x(R) = x_1 = r_0 (\mathrm{mod}\ n)$, the ECDSA signature ($r_0$, s_0) will be accepted.

3.2 Generate and Verify SM2-ECDSA

3.2.1 Generate keys of SM2-ECDSA

For generating the signature keeping with the the Chinese digital signature standard SM2 (SM2-ECDSA), the first phase is to generate the public key Qs = (Qx, Qy) and the secret key ds. The proposed system will perform the following steps:

1) Select a random number ds $\in$ [1, ..., n − 1].

2) Compute $Q_s = d_s P$.

3) Return (Qs, ds), where Q is the public key and d is the secret key.

3.2.2 Generate SM2-ECDSA

Then, generate the SM2-ECDSA signature using the secret key ds:

Let $Z = H(\mathrm{ENT\ L}||\mathrm{ID}||a||b||Qx||Qy)$.

ENT L represents the length of ID.

ID denotes the identity of the owner.

a and b are the parameters in E : $y^2 = x^3 + ax + b$.

Qx and Qy denotes the x-coordinate and y-coordinate of public key Q respectively.

2) Let $m = Z||m$.

3) Compute the hash value of message m as: $e = h(m)$. h is the selected hash function.

4) Select a random number k $\in$ [1, ..., n − 1]. k is called a nonce.

5) Compute $R = (x, y) = kP$.

6) Compute $r = e + x \bmod n$, where x is the x-coordinate of R in the above steps. If $r = 0$ or $r + k = n$, regenerate the random number k.

7) Compute $s = (1 + d_s)^{-1}(k - rd_s) \bmod n$. If $s = 0$, regenerate the random number k.

8) Return (r, s) as the SM2-ECDSA signature.

3.2.3 Verify SM2-ECDSA

To verify the SM2-ECDSA signature (r', s') on a message m', the system will perform the following steps:

1) Check whether $r' \in [1, \ldots, n - 1]$, if not, return false and exit.

2) Check whether $s' \in [1, \ldots, n - 1]$, if not, return false and exit.

3) Let $m_0 = Z||\, m'$.

4) Compute $e' = h(m')$.

5) Compute $t = (r' + s') \bmod n$. If t = 0, output 0 and exit.

6) Compute $(x_1', y_1') = s'P + tQ$.

7) Compute $R = (e' + x_1') \bmod n$.

8) If $R = r'$, return true; else return false.

4 Comparison

In this section, we will analyze and compare the performance and security of ECDSA and ECDSA-SM2 in terms of time consumption and security.

In general, the implementation of the two algorithms is similar to some extent. In addition, the security of ECDSA-SM2 and ECDSA are at the same level, but the security of ECDSA-SM2 is higher than that of ECDSA.

4.1 Security

1) The security of ECDSA and ECDSA-SM2 are both based on the difficulty of solving the discrete logarithm problem of the ellipse curve (ECDLP).

2) The elliptic curve it relies on can use the recommended curve such as NIST, but SM2 does not give a recommended curve. The generation of curve parameters needs to be generated by a certain algorithm.

3) Besides, in the phase of generating signatures, both ECDSA and ECDSA-SM2 generate a random number k. Since the attacker does not know k and the secret key d which is computationally infeasible to learn k and d by solving two elliptic curve discrete logarithm problems, the security of ECDSA-SM2 and ECDSA are at the same level.

4) In ECDSA, there is no preprocessing of the signed message m, while in the ECDSA-SM2, it is necessary to let $Z = H(ENT\ L||ID||a||b||G||Y)$ and $m' = Z||m$. ENT L represents the length of ID. ID denotes the identity of the owner. a and b are the parameters in E. Qx and Qy denotes the x-coordinate and y-coordinate of public key Q respectively.

The users own information, curve parameters, base points, and users public key point information are concatenated in Z. Hence, the processed message m_0 includes the signer's individual information, which improves the security of users and provides stronger anti-repudiation as well.

4.2 Time consumption

We compared the time consumption of generating 3000 signatures for the SM2-ECDSA, ECDSA and RSA2048 respectively.

The experiment environment used for implementing the simulations is windows 10 with an Intel Core i7-8750h CPU @2.20GHz and 16 GB memory.

The selected curve of ECDSA is secp256k1 that Bitcoin adopted. The hash function that SM2 chooses is SM3 while ECDSA and RSA choose SHA256. The selected typical RSA algorithm is RSA2048. Fig 2 describes the time consumption comparison result with the increasing message size from 2KB to 256KB. Obviously, when the message size is small such as 2KB to 32KB, the time consumption of ECDSA-SM2 and ECDSA are greatly lower than that of RSA2048. When the message size reaches nearly 256KB, the overhead of ECDSA-SM2 is similar to that of ECDSA and still much smaller than the time consumption of RSA2048.

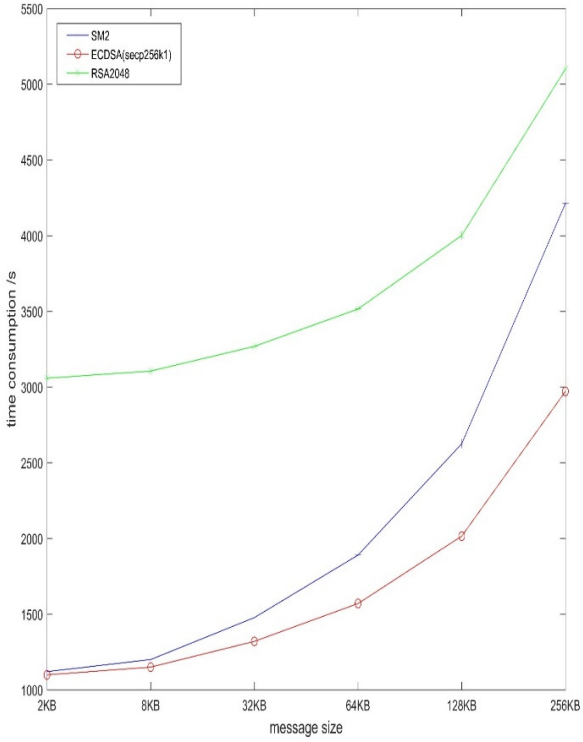

Figure 4: The time consumption

5 Conclusion

This paper put forwards a security-enhanced blockchain system. The proposed system supports standard ECDSA signature and ECDSA-SM2 simultaneously for the security requirement. As far as we are concerned, it is the only protocol that implements ECDSA and ECDSA-SM2 in the blockchain system simultaneously. In future research, to enhance the anonymity of blockchain, the blind signature or ring signature that compatible with ECDSA is a meaningful topic and direction in the blockchain.

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